Hyponatremic Seizure

Section I: Scenario Demographics

Scenario Title:	Hyponatremic Seizure
Date of Development:	23/05/2015 (DD/MM/YYYY)
Target Learning Group:	☐ Juniors (PGY 1 – 2) ☐ Seniors (PGY ≥ 3) ☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to a relatively rare presentation that requires critical intervention and can masquerade as other common conditions.
CRM Objectives:	Communicate clearly with team members and ask for advice if unsure of dosing.
Medical Objectives:	1) Demonstrate an appropriately broad initial workup for the elderly patient who is generally unwell. 2) Recognize hyponatremia as a cause of seizure. 3) Treat hyponatremic seizure with hypertonic saline.

Case Summary: Brief Summary of Case Progression and Major Events
A 93 year old woman comes in with family. They are concerned about general weakness, worsening PO intake over the last few months, and new confusion. As the team takes a history and starts the initial workup, the patient will begin to seize. She will seize continuously until hypertonic saline or a paralytic is given. After two doses of benzodiazepine, a critical result showing severe hyponatremia will come back. The team is expected to administer hypertonic saline, which will stop the seizure. The patient will remain somnolent after this dosing, and as the team prepares to intubate, she will seize again, requiring a repeated dose of hypertonic saline.

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.

Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient	*Select most important dimension(s)*	☒ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A

Confederates	Brief Description of Role
Patient's daughter	Fill in history of last few months. Will become quite panicky when patient starts seizing and will need to be taken to the side.
Nurse	Will cue team to seizures and remind them repeatedly that patient is still seizing

B. Required Monitors		
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

C. Required Equipment		
☒ Gloves	☒ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit
☐ Blood Products	☒ ET Tubes	☐ Other:
☐ Intraosseous Set-up	☐ LMA	☐ Other:

D. Moulage
None required.

E. Approximate Timing					
Set-Up:	3 min	Scenario:	15 min	Debriefing:	15 min

Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case
Agnes Jones is a 93 year old female who has been brought to the ED by her daughter. The family has noticed that Agnes is not eating well over the last few months. She seems weak. Now, over the last day or so, she seems confused.

B. Patient Profile and History

Patient Name: Agnes Jones	Age: 92	Weight: 50kg
Gender: ☐ M ☒ F	Code Status: Full	
Chief Complaint: "Generally weak"		
History of Presenting Illness: **Poor PO intake over months.** Has been weak, and getting out of bed less and less. **Today, she has been confused**. She doesn't seem to know where she is and isn't making sense. She's never been like this before.		

Past Medical History:	**HTN**	Medications:	**HCTZ 25mg daily**
	Hypothyroidism		**ASA 81mg daily**
	Depression		**Levothyroxine 125mcg daily**
			Celexa 20mg daily
Allergies: Sulfa			
Social History: **Lives with daughter. A few months ago, was very active.** Non-smoker. No EtOH.			

Review of Systems:	CNS:	**Confusion over the last day.**		
	HEENT:	Nil.		
	CVS:	No CP, no palpitations.		
	RESP:	No SOB.		
	GI:	No V/D. Last BM yesterday.		
	GU:	No UTI symptoms.		
	MSK:	**Complains of feeling weak. Standing less and less**	INT:	Thin skin.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display	
HR: 95/min	BP: 90/60	RR: 12/min	O_2SAT: 96% RA
Rhythm: NSR	T: 36.4^oC	Glucose: **6.4** mmol/L	**GCS: 13 (E 4 V4 M5)**
General Status: **Looks cachectic.**			

CNS:	**Confused. Not oriented to time or place. Not obeying commands.**		
HEENT:	**No signs HI. PERLA. 3mm.**		
CVS:	No murmur.		
RESP:	GAEB, no advent.		
ABDO:	Soft, NT.		
GU:	Nil.		
MSK:	**Not cooperating with exam, but 3-4/5 strength**	SKIN:	**Thin skin.**

Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: NSR HR: 95/min BP: 90/60 RR: 12/min O_2SAT: 96% RA T: 36.4°C	Confused. Not oriented. Looks frail and cachectic.	<u>Learner Actions</u> -☐ IV access, monitors, bolus -☐ Cap sugar: 6.4 -☐ Blood work: trop, lytes, extended lytes, TSH -☐ Order urine C+S, CXR -☐ ECG -☐ More history, px exam	<u>Modifiers</u> *Changes to patient condition based on learner action* - 1L NS → BP 100/60 <u>Triggers</u> *For progression to next state* - 3 min → **2. Seizure**
2. Seizure HR →140 BP → 155/95 O_2SAT → slowly decrease to 86%	Patient having generalized, tonic, clonic seizure.	<u>Learner Actions</u> -☐ Check cap sugar (if not done previously): 6.4 -☐ Administer benzo x2 doses -☐ Consider phenytoin -☐ Consider toxicologic causes -☐ Prepare for intubation	<u>Modifiers</u> - No agents will stop seizure <u>Triggers</u> - 2 doses benzodiazepine given → **3. Blood work back** - 6 min → **3. Blood work back**
3. Blood work Back Same as previous state.	Patient still seizing. **Team handed critical result at onset of state.**	<u>Learner Actions</u> -☐ Administer 3% NS 100ml over 10 minutes -☐ Send urine osmols, serum osmols, TSH (if not yet done) -☐ Insert foley	<u>Modifiers</u> - No hypertonic given → patient continues seizing <u>Triggers</u> - Hypertonic given → **4. Somnolent** - 10 min → **4. Somnolent**
4. Somnolent HR → 100 BP → 95/65 O2 → 91% RR → 6	Patient not seizing, but making poor respiratory effort, not responsive.	<u>Learner Actions</u> -☐ Prepare for intubation	<u>Modifiers</u> - No plan to intubate → O2 87% <u>Triggers</u> - Prepping for intubation → **5. Repeat seizure** - 12 min → **5. Repeat seizure**
5. Repeat seizure HR →140 BP → 155/95 O_2SAT → slowly decrease to 86%	Patient has another GTC seizure	<u>Learner Actions</u> -☐ Administer repeat dose of 3% NS 100 ml over 10 minutes -☐ Prepare for intubation	<u>Modifiers</u> - Hypertonic given → stop seizure <u>Triggers</u> Intubation → **6. Intubation** 15 min → **END CASE**
6. Intubation O2 → 93% with pre-oxygenation Once paralytic given: RR → 0 → 12 (vented) HR → 110 BP → 110/75	Patient non-responsive	<u>Learner Actions</u> -☐ Pre-oxygenate -☐ RSI with paralytic -☐ Post-intubation sedation -☐ Post-intubation CXR -☐ Call for CT head -☐ Call ICU -☐ Order repeat lytes STAT -☐ ± EEG to monitor for seizure	<u>Modifiers</u> <u>Triggers</u> - Successful intubation and post-intubation management started→ **END CASE**

Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results
Verbal report received re: critical result → Na 101

Images (ECGs, CXRs, etc.)	
Pre-intubation CXR (http://radiologypics.com/2013/01/25/normal-female-chest-radiograph/)	ECG showing NSR (http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/12/normal-sinus-rhythm.jpg)
Post-intubation CXR (https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg)	

Section VIII: Debriefing Guide

General Debriefing Plan			
☐ Individual	☒ Group	☐ With Video	☒ Without Video

Objectives	
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Sample Questions for Debriefing
1) Did you consider hyponatremia as a cause of seizure before the blood work was back? 2) What is the dose of hypertonic saline in a critical hyponatremic patient? What are the indications for hypertonic saline? 3) How did the team problem solve the ongoing seizures? Was the team leader open to suggestions? What are some techniques a leader can use to ensure the team feels comfortable contributing their thoughts?

Key Moments
Initial work-up
Management of seizure
Responding to critical blood result by treating with hypertonic saline